

Environmental recurrence and directional cache information select a costly memory-use allele without a resolved one-bit population benefit

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Abstract

Prior theory and experiments establish that environmental predictability can favor memory-like inheritance, learning and anticipatory effects, and that memory can carry costs. We asked narrower, prospectively specified questions in an abstract computational model, in two prospective studies. Does lag-two environmental recurrence enrich a mutable allele for using a supplied one-record private cache when each retrieval displaces one of two fresh exploratory proposals; does the directional information in the retrieved record, rather than the mere loss of a fresh proposal, drive that selection; and does either coincide with a meaningful gain in population accuracy? In a finite population of 32 haploid searchers, a binary use policy mutated symmetrically at rate $1/32$ and was selected only through the objective scores of the candidates it caused to be proposed. Label-blind SHAM arms supplied an exact neutral benchmark for the label process. In Study 001, across 41,600 independent paired blocks (332,800 paths and 10,905,190,400 objective queries), late-window start contrasts were bounded within one expected individual (95% familywise) over the 256-update horizon, passing the prespecified start-state gate, and recurrence strongly enriched the memory-use allele. Relative to the neutral $1/2$, the start-averaged excess was 0.310292 (95% familywise Hoeffding-Bonferroni interval [0.302482, 0.318102]); relative to the nonrecurrent law it was 0.441635 ([0.426015, 0.457256]). By contrast, the recurrent-law ACTIVE minus SHAM accuracy contrast was 0.009377 ([-0.005138, 0.023892]), bounded below the prespecified one-correct-bit scale of 0.03125, with sign unresolved, and the recurrence interaction in accuracy, 0.011360 ([-0.017669, 0.040390]), was unresolved. Under the nonrecurrent law, ACTIVE accuracy was descriptively 0.001983 below SHAM. Study 002, a post-v1.0 prospective extension with a new cohort of 41,600 blocks (249,600 paths and 8,178,892,800 objective queries), kept only the aligned recurrent law and compared directional cache retrieval (INFO) with a recurrence-gated control that preserves each retrieved candidate's distance to its parent but scrambles its direction (NONINFO). Both start contrasts again passed the gate. Directional information raised late-window memory-use frequency by 0.184203 ([0.169688, 0.198718]) relative to NONINFO, which itself exceeded the neutral $1/2$ by 0.126083 ([0.118825, 0.133340]). The frozen primary estimand, the total population-accuracy effect of directional information relative to NONINFO, was 0.008650 ([-0.006521, 0.023821]): bounded below the positive one-bit scale, wholly inside $\pm 1/32$ and with sign unresolved. Under one deliberately aligned recurrence law and one fixed opportunity cost, directional cache information therefore strongly amplified selection on an information-use policy, while its total effect on population performance remained bounded below one correct bit. The results do not address the origin of memory architecture, other recurrence laws, other costs or biological systems.

1. Introduction

Recurring environments make the past informative about the future, and a substantial literature shows that this can favor carrying information across time. Kuijper and Johnstone found that environmental stability and predictability can favor positive parental effects and long-term phenotypic memory under genetically determined transmission rules [1]. Carja and Plotkin showed that, in periodically fluctuating environments, an intermediate degree of phenotypic memory maximizes the fixation probability of an allele that permits heritable phenotypic heterogeneity [2]. Dey, Proulx and Teotónio showed experimentally that temporal environmental structure can select maternal effects [8]; Skanata and Kussell derived evolutionary regimes for response networks with phenotypic memory in random environments [9]; and Kronholm found that anticipatory effects mediated by epigenetic changes evolve when an environmental cue is reliable, the environment changes at intermediate rates and the costs of the effect are low [7].

Models of learning reach related conclusions. Wright and colleagues allowed memory weighting and current sampling effort to evolve under explicit costs and found that memory increased with temporal autocorrelation and interacted with costly sampling [3]. Arehart and Adler quantified the costs and benefits of memory duration and exploration in a minimal learning model and found that the marginal benefit of learning is often small [4]. Morgan, Suchow and Griffiths inserted human participants as agents into evolutionary simulations with simulated loci for learning and memory and found that rapid environmental change dampened their evolution [5]. Among the studies identified by our bounded search, this is a close precedent for an evolving memory-use trait, although its learners were human participants rather than a cleanly mutable allele. Other work is at least as close on other dimensions: Carja and Plotkin analyzed a mutable modifier allele and its fixation in a finite population [2], and Grabowski and colleagues studied the early evolution of memory usage in digital organisms [6].

The value and cost of memory that is already present are also well studied. Dunlap and Stephens linked environmental change to optimal sampling and memory weighting [10]. Lambert and Kussell measured bacterial memory under fluctuating carbon sources and modeled its long-term fitness benefit [11]. Rescan and colleagues linked phenotypic memory to population growth and extinction risk in a noisy environment [12]. George optimized memory-driven phenotypic adaptation in fluctuating environments [13], and Jain, Jolly and George added adaptation costs [14]. Pollack, Nozoe and Kussell analyzed selection on gene regulation and phenotypic memory under metabolic fluctuations [15]. Leadbeater and Watrobska emphasize that memory can be shaped by selection along many mechanistic dimensions rather than as a single scalar trait [16].

Against this background, we do not claim to show that memory evolves, that predictability favors memory, or that costly memory can be selected; those propositions are occupied by the work above. The contribution here is a specific joint design, executed with a new implementation and new random cohorts, that allows two outcomes to be judged separately. First, the memory architecture, a one-record private cache, is supplied, and only a binary use policy mutates and is selected, in a finite population with explicitly stochastic survival. Second, the cost of using memory is a one-for-one search-budget opportunity cost: a carrier that retrieves its cache gives up one of its two fresh proposals, with no added fitness penalty or tunable coefficient. Third, a label-blind SHAM arm supplies an exact neutral benchmark for the label process at the same query budget. Fourth, allele enrichment and population accuracy were judged by prospectively separate decision rules, each with a stated meaningful scale. Fifth, a recurrence-gated, parent-distance-matched control separates the directional content of a retrieved record from the loss of the fresh proposal it displaces.

The separation matters because the two outcomes need not move together. An allele can rise in frequency because it improves its carriers' chances of being among the survivors while the population's average fit to the environment changes little. Carriers may displace non-carriers that would have achieved similar accuracy, or gains at recurrent updates may be offset by lost exploration at non-recurrent updates. Treating allele enrichment as evidence of population benefit would conflate these possibilities.

We report the accepted results of two prospective studies of the same model. In Study 001 (PHASE2-MUTABLE-MEMORY-001), under one recurrent target law that was deliberately aligned with the cache delay, lag-two recurrence strongly enriched the mutation-generated memory-use allele above both its exact neutral expectation and its frequency under a nonrecurrent control, despite the fixed opportunity cost. The corresponding ACTIVE minus SHAM accuracy contrast was bounded below the prespecified one-correct-bit scale, with sign unresolved, and the recurrence interaction in accuracy remained unresolved. Study 001 could not say how much of that selection responded to what the cache contained rather than to the fresh proposal that retrieval displaced. Study 002 (PHASE2-PERFORMANCE-CONVERSION-002), a post-v1.0 prospective extension with a new random cohort, addressed that question within the aligned law by comparing directional retrieval (INFO) with a recurrence-gated control that keeps each retrieved candidate at the cache's distance from its parent but scrambles its direction (NONINFO). Directional information strongly increased selection on the memory-use policy, yet its total effect on population accuracy was bounded below the one-bit scale with sign unresolved. We present the two studies as a single result: in this model, recurrence selected a costly memory-use policy, directional cache information strongly amplified that selection, and neither was accompanied by a demonstrated commensurate population-performance advantage at the fixed scale.

2. Model and prospective design

This paper reports two prospective studies of one model. Sections 2.1-2.7 describe Study 001, and Section 2.8 describes the Study 002 extension, which changes only the construction of one candidate. For each study, the specification, decision table, cohort size, audit plan and claim limits were frozen before any production outcome was generated. For Study 001, a non-implementing AI reviewer returned a terminal GO review of the frozen specification before production was authorized; the specification hash, review record and timestamps are given in Section 7. In neither study was any parameter, cohort, horizon, target law, mutation rate or cost changed after outcomes were observed. The complete update order is given in Supplementary Sections S2-S7 for Study 001 and S17-S18 for Study 002.

2.1 Population, search state and objective

The population contains 32 haploid individuals in fixed slots 0-31. Each individual carries an unsigned 32-bit genotype and a binary policy allele, M (use the private cache) or F (fresh exploration). An M individual also holds a private cache that is either invalid or contains one 32-bit genotype; F individuals hold no valid cache. The allele jointly determines whether the one-step record is retained and whether it is retrieved; every individual has a cache slot, but only M carriers receive a valid record, so this design cannot separate selection on retention from selection on retrieval. Lineage identifiers are audit metadata only and are never read by any scientific operator. Each path runs for updates 1-256. At update t the environment presents a 32-bit target T_t , and the raw objective of a genotype is its Hamming mismatch to T_t . Population accuracy at an update is one minus the mean survivor mismatch divided by 32, measured after survival; a population of uniformly random genotypes has expected accuracy $1/2$. The late window is updates 193-256 inclusive.

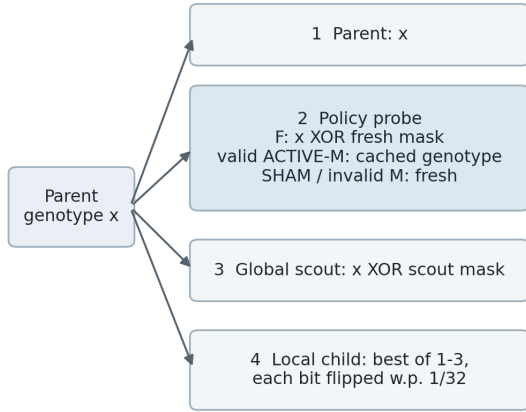
The population size, bit length, four-candidate architecture, lag-two law and selection weights were inherited from earlier internal private-memory studies so that the changed question, selection on a mutable use policy, would be identifiable. The implementation and all random cohorts are new and import no producer code from those studies. This study is not a replication of them.

2.2 Candidate budget and the one-proposal opportunity cost

Every parent produces exactly four evaluated candidates per update, so every update makes exactly 128 objective queries regardless of allele, arm or law. The candidates are (1) the unchanged parent; (2) a policy probe; (3) a global scout, the parent XOR an independent uniform 32-bit mask; and (4) a local child, made by taking the lowest-mismatch member of candidates 1-3 (ties broken by a keyed random key and then by family order) and flipping each of its 32 bits independently with probability $1/32$. For an F individual the policy probe is the parent XOR a second independent uniform mask, so an F carrier makes two fresh, effectively uniform proposals per update. For a valid M carrier in the ACTIVE arm, the policy probe is the cached genotype itself. Because the probe is also eligible to be the donor of the local child, a useful cache can shape two of the four candidates.

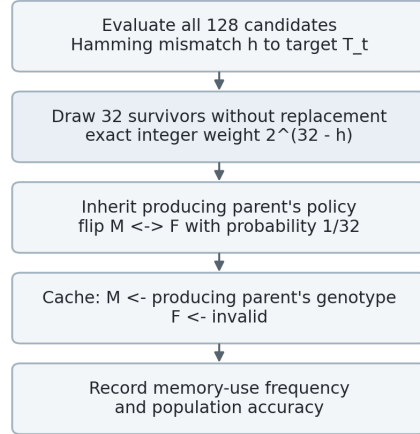
The cost of memory is therefore an opportunity cost on a fixed search budget: retrieval replaces exactly one of two effectively uniform fresh proposals with the cached record. There is no score penalty and no tunable cost coefficient. This is a model-specific search-budget trade-off, not a calibrated energetic, metabolic or biological cost, and its magnitude is conditional on the frozen fresh-probe distribution. When the cache is uninformative, an M carrier loses one fresh draw at the same query count. When the cache closely matches the current target, retrieval supplies a candidate that a uniform fresh draw is unlikely to match. Genotypes are not deduplicated: if the cache equals the unchanged parent, both copies are separately evaluated and the update still makes 128 queries.

(a) Four candidates per parent per update



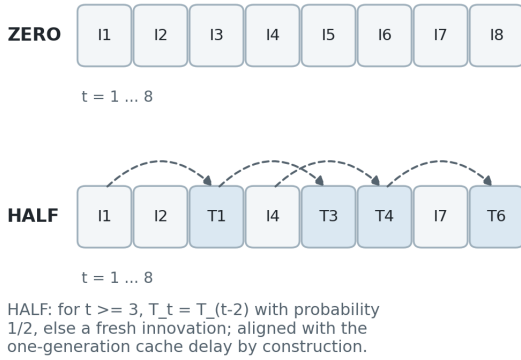
Retrieval replaces one of two uniform fresh proposals; 128 objective queries per update in every cell.

(b) Population event order at update t



No label, cache or lineage identifier enters selection.

(c) Target laws (illustrative, not data)



(d) Paired cells and inferential families

Each block: 2 arms (ACTIVE, SHAM) x 2 laws (ZERO, HALF) x 2 starts (ALL_F, ALL_M) = 8 paired cells sharing all keyed random values. SHAM never reads labels or caches: exact neutral label benchmark.

Primary family ($\alpha_{\text{each}} = 0.05/4$): allele $C_{\text{abs}} = A_{\text{HALF}} - 1/2$; $C_{\text{rec}} = A_{\text{HALF}} - A_{\text{ZERO}}$ Start-state gate: $D_{\text{HALF}}, D_{\text{ZERO}}$ inside $(-1/32, +1/32)$

Secondary family ($\alpha_{\text{each}} = 0.05/2$): accuracy $P_{\text{abs}} (\text{HALF: ACTIVE} - \text{SHAM})$; $P_{\text{rec}} (\text{interaction})$ Scale: one correct bit per survivor $(1/32)$; cannot alter the primary decision

Figure 1. Model and prospective design. (a) One update for one parent: four evaluated candidates, of which only the policy probe depends on the allele and arm; a valid ACTIVE-M carrier replaces one of its two fresh uniform proposals with its cached genotype. (b) Population-level event order: evaluation of 128 candidates against the current target, exact integer-weighted sequential sampling of 32 survivors, symmetric policy mutation, and cache assignment. (c) The two target laws; HALF returns to the target from two updates earlier with probability 1/2, which aligns with the one-generation cache delay by construction. (d) The eight paired cells per block and the separate primary (allele) and secondary (accuracy) inferential families. The schematic contains no outcome values.

2.3 Target laws and the deliberate HALF alignment

For every block and update, an innovation I_t (a uniform 32-bit word) and a fair recurrence bit R_t are generated whether or not they are used. Under ZERO, $T_t = I_t$ at every update. Under HALF, $T_1 = I_1$ and $T_2 = I_2$; for $t \geq 3$ the target returns to T_{t-2} when $R_t = 1$ and otherwise equals I_t . Non-recurrent HALF updates use the same innovation as ZERO at that update.

The cache of a survivor records its producing parent's genotype from the update in which the survivor was made, and that parent had itself survived selection against the previous target. A cache used at update $t+1$ therefore holds a genotype that survived selection against T_{t-1} , which under HALF is the target at update $t+1$ with probability 1/2. HALF was a prior, outcome-informed choice made precisely to align the one-generation cache delay with a two-update target return; it favors retrieval by construction. No other lag, copy probability, target

law, cost or model class was tested.

2.4 Selection, policy mutation and cache inheritance

All 128 candidates are evaluated. Candidate j with mismatch h_j receives the exact integer weight $2^{(32-h_j)}$, proportional to $2^{(-h_j)}$. Thirty-two survivors are drawn sequentially without replacement, each with probability proportional to its weight among the candidates not yet selected. This is an exact Plackett-Luce weighted sample implemented entirely in unsigned integer arithmetic, with unbiased bounded integers produced by multiply-high rejection (Supplementary Section S5). No label, cache, lineage identifier or penalty enters selection; the policy allele affects survival only through the genotypes it causes to be proposed.

Each survivor first inherits its producing parent's policy. Then, independently in each survivor slot, the policy flips $M \leftrightarrow F$ with probability $\mu = 1/32$. After mutation, an F survivor has an invalid cache and an M survivor's cache becomes its producing parent's pre-update genotype. From update 2 onward every M carrier therefore holds a valid one-step lineage cache. A new $F \rightarrow M$ mutant first uses its record on the next update, and an $M \rightarrow F \rightarrow M$ history carries no older hidden record. Allele frequencies and accuracy are recorded after this transition.

2.5 Paired cells and the exact SHAM benchmark

Each of 41,600 independent blocks contains the same eight paired cells: information arm (ACTIVE or SHAM) $\times$ target law (ZERO or HALF) $\times$ policy start (ALL_F or ALL_M). Within a block, all eight cells share 32 initial uniform genotypes and all coordinate-keyed random values; caches start invalid. Random values come from counter-based Philox4x32-10 streams indexed by purpose, block, update, entity and subindex, and every draw is generated whether or not it is used, so no label, branch, arm or law changes any later random value (Supplementary Section S7).

In SHAM, the policy probe is always the fresh construction, so genotypes never depend on a label or cache. Two exact identities follow. N1: SHAM genotype, mismatch, accuracy and survivor trajectories are bit-identical between the ALL_F and ALL_M starts. N2: SHAM labels in the two starts are exact complements after every update, so their memory-use frequencies sum to 1 at every update and their start-averaged late-window frequency is exactly 1/2 in every block. With symmetric mutation and $r = 1 - 2\mu = 15/16$, the SHAM expectations are $P(M \text{ at } t \mid \text{ALL_F}) = [1 - r^t]/2$ and $P(M \text{ at } t \mid \text{ALL_M}) = [1 + r^t]/2$. The expected ALL_M minus ALL_F difference is $r^{256} = 6.678005283720385e-08$ at update 256 and $r^{193}(1 - r^{64})/(64(1 - r)) = 9.57983820600146e-07$ averaged over the late window.

SHAM is an exact neutral benchmark for the label process (N1, N2; expected start-averaged frequency 1/2) and a paired control for query budget and keyed random draws. Because ACTIVE genotypes and genealogies diverge from SHAM at the first valid-M retrieval, SHAM does not hold realised drift or genotype state fixed for ACTIVE paths; ACTIVE-SHAM contrasts are arm-level comparisons within the model, not path-level counterfactuals. It does not by itself separate the information carried by the cache from the mere loss of one fresh proposal; the ZERO law, under which the cache carries no lag-two information, supplies that contrast through C_{rec} , and Study 002 (Section 2.8) supplies a within-law contrast. It does not represent any alternative cost, retrieval rule or memory architecture. Its neutral formulas justify the late window in expectation for SHAM only and do not establish that ACTIVE paths have converged.

2.6 Estimands, intervals and frozen decisions

For each block, let $L(\text{arm}, \text{law}, \text{start})$ be the mean memory-use frequency over updates 193-256. Define $A_HALF = [L(\text{ACTIVE}, \text{HALF}, \text{ALL_F}) + L(\text{ACTIVE}, \text{HALF}, \text{ALL_M})]/2$ and A_ZERO analogously. The four primary estimands are:

- $C_abs = A_HALF - 1/2$, the recurrent-law excess over the exact neutral expectation;
- $C_rec = A_HALF - A_ZERO$, the excess attributable to recurrence;
- $D_HALF = L(\text{ACTIVE}, \text{HALF}, \text{ALL_M}) - L(\text{ACTIVE}, \text{HALF}, \text{ALL_F})$, the start-state contrast under HALF; and
- $D_ZERO = L(\text{ACTIVE}, \text{ZERO}, \text{ALL_M}) - L(\text{ACTIVE}, \text{ZERO}, \text{ALL_F})$, the start-state contrast under ZERO.

Let $P(\text{arm}, \text{law})$ be start-averaged late-window mean population accuracy. The two secondary estimands are:

- $P_abs = P(\text{ACTIVE}, \text{HALF}) - P(\text{SHAM}, \text{HALF})$, the recurrent-law accuracy contrast; and
- $P_rec = [P(\text{ACTIVE}, \text{HALF}) - P(\text{SHAM}, \text{HALF})] - [P(\text{ACTIVE}, \text{ZERO}) - P(\text{SHAM}, \text{ZERO})]$, the recurrence interaction in accuracy.

The meaningful allele scale is $\Delta = 1/32 = 0.03125$, one expected individual in the 32-member population; it refers to a late-window expected excess, not to the persistence of one named carrier. The meaningful performance scale is $\Delta_P = 1/32$, one additional correct bit per survivor on average, which is 3.125 percentage points of accuracy.

Each family is controlled at 95% familywise coverage by two-sided Hoeffding intervals with a Bonferroni split. For a block variable with range length R and $n = 41,600$ independent blocks, the half-width is $h = R \sqrt{[\ln(2/\alpha_each)]/(2n)}$. In the primary family, $\alpha_each = 0.0125$; C_abs has range length 1 and half-width 0.00781023, and C_rec , D_HALF and D_ZERO have range length 2 and half-width 0.01562046, just below $\Delta/2 = 0.015625$. In the secondary family, $\alpha_each = 0.025$; P_abs has range length 2 and half-width 0.01451463, and P_rec has range length 4 and half-width 0.02902925. No empirical variance, normal approximation, within-path independence, pilot or favorable pairing enters these bounds. The two families are controlled separately, not jointly.

The primary decision table was applied in order: (1) failure of any deterministic identity, seed or counter audit, saved-output reconstruction or query count gives INVALID; (2) if either D_HALF or D_ZERO has an interval not wholly inside $(-\Delta, +\Delta)$, the result is START-DEPENDENT and the question unresolved; (3) if the lower bounds of both C_abs and C_rec exceed Δ , the result SUPPORTS RECURRENCE-ATTRIBUTABLE SELECTIVE ENRICHMENT; (4) if the lower bound of C_abs exceeds Δ and the upper bound of C_rec is at or below Δ , enrichment is not attributable to recurrence at the fixed scale; (5) if the upper bound of C_abs is at or below Δ , the result is a bounded negative at the one-individual scale; (6) every other pattern is unresolved. Each secondary estimand was classified, in order, as meaningful positive performance if its lower bound exceeds Δ_P , meaningful adverse performance if its upper bound is below $-\Delta_P$, BOUNDED BELOW THE POSITIVE ONE-BIT SCALE if its upper bound is at or below Δ_P , and UNRESOLVED otherwise. Secondary classifications cannot alter the primary decision.

Sixteen absolute cell means, the late-window memory-use frequency and accuracy for each of the eight cells, were saved as descriptive records without inferential bounds. Together with the six inferential records they form the 22 prespecified saved estimate records. The descriptive means support interpretation only.

2.7 The start-state gate, and why allele frequency is not accuracy

The start-state gate, rule 2, asks whether populations started entirely from F and entirely from M reach the same late-window ACTIVE frequency to within one expected individual. If they did not, the late window would partly reflect where paths began rather than a late-window state approached from both starts within the frozen horizon, and no enrichment classification would be made. The gate is applied before any enrichment rule. Passing it shows only that both late-window start contrasts were bounded within $(-1/32, +1/32)$ at 95% familywise coverage over the 256-update horizon; it does not demonstrate start-independence in general, stationarity or a unique long-run state.

Allele frequency measures which policy the survivors carry; accuracy measures how well the survivors match the target. The two are linked only through the genotypes that M carriers propose. A policy can raise its carriers' representation among survivors by supplying high-scoring candidates at recurrent updates while costing a fresh proposal at every update, so its effect on mean survivor accuracy can be smaller than, or differently signed from, its effect on its own frequency. This is why the two outcomes were assigned separate estimands, scales and decision rules in advance.

2.8 Study 002: separating directional cache information from displacement

Study 002 retains every Study 001 operator: the population, genotype, mutation rate $1/32$, cache inheritance, four-candidate budget with 128 objective queries per update, exact integer-weighted survival, 256-update horizon and late window. It uses only the HALF law, which remains deliberately aligned to the cache delay; there is no ZERO arm. It changes only the policy probe, in three arms. For a parent with genotype x and valid cache c :

- **INFO:** a valid M carrier proposes c , exactly as in the Study 001 ACTIVE arm;
- **NONINFO:** a valid M carrier proposes c at innovation updates ($t < 3$ or $R_t = 0$), and at recurrent updates ($t \geq 3$ and $R_t = 1$) proposes $x \text{ XOR } \pi_{(b,t)}(c \text{ XOR } x)$, the cache displacement with its bit positions moved by a shared permutation; and
- **SHAM:** every label uses the fresh construction, as in Study 001.

F carriers and invalid-cache M carriers use the fresh construction in every arm. The experimental operator reads R_t ; the evolving policy cannot. INFO and NONINFO therefore follow identical rules at innovation updates and differ only in whether a valid cache displacement keeps its direction at recurrent updates. Both pay the same one-proposal displacement cost at every valid-M use.

$\pi_{(b,t)}$ is one exact uniform random permutation of the 32 bit positions, generated for every block b and update t regardless of branch or arm by a Fisher-Yates shuffle with unbiased multiply-high rejection, and shared by all cells and all parents of that block and update (Supplementary Section S18). Because permuting bit positions preserves Hamming weight, a NONINFO decoy lies exactly as far from its parent as the cache does. The shared permutation also preserves overlaps among the displacement masks of different parents under one common relabeling of coordinates, but it does not preserve every cross-genotype relation; for example, siblings that share one cached genotype do not share one decoy endpoint. A displacement of weight 0 or 32 cannot be scrambled and retains its direction; these cases are counted in the diagnostics.

Each of 41,600 new independent blocks contains six paired cells, arm (INFO, NONINFO, SHAM) $\times$ policy start (ALL_F, ALL_M), sharing 32 initial uniform genotypes and every keyed random value at compatible coordinates, including the permutations. Random values come from the same Philox4x32-10 construction under a new namespace; no genotype, target, trajectory, block,

random tape or outcome from Study 001 was reused. The Study 002 producer extends the authenticated Study 001 producer, and its unchanged operators were required to remain fixture-identical. SHAM obeys the same identities N1 and N2 as in Study 001, so its start-averaged late-window memory-use frequency is exactly 1/2 in every block; it serves both as the neutral allele benchmark and as the no-memory performance reference.

For block b , let $P(\text{arm}, \text{start})$ be the late-window mean population accuracy and $L(\text{arm}, \text{start})$ the late-window mean memory-use frequency, and let P_{arm} and A_{arm} be their equal-weight averages over the two starts. The primary family contains three estimands:

- $\Delta_P = P_{\text{INFO}} - P_{\text{NONINFO}}$, the total population-accuracy effect of directional cache information beyond the parent and matched displacement length, mediated through evolved policy use; and
- $D_{\text{INFO}} = L(\text{INFO}, \text{ALL}_M) - L(\text{INFO}, \text{ALL}_F)$ and $D_{\text{NONINFO}} = L(\text{NONINFO}, \text{ALL}_M) - L(\text{NONINFO}, \text{ALL}_F)$, the finite-horizon start-state gate.

INFO and NONINFO share policy rules but not realized policy states, so Δ_P includes any divergence in policy frequency, cache use, survivor composition and descendants. It is a total effect, not a per-use value-of-information estimate. Two secondary families cannot alter the primary decision. The allele family contains $E_{\text{INFO}} = A_{\text{INFO}} - A_{\text{NONINFO}}$, the allele-frequency effect of cache direction, and $E_{\text{NONINFO}} = A_{\text{NONINFO}} - 1/2$, selection under the matched control relative to exact SHAM neutrality. The no-memory performance family contains $B_{\text{INFO}} = P_{\text{INFO}} - P_{\text{SHAM}}$ and $B_{\text{NONINFO}} = P_{\text{NONINFO}} - P_{\text{SHAM}}$.

The meaningful scale is again 1/32: one correct bit in mean accuracy (3.125 percentage points) or one expected individual in allele frequency. Intervals are two-sided Hoeffding-Bonferroni intervals over $n = 41,600$ blocks. The primary family uses $\alpha_{\text{each}} = 0.05/3$; each primary block variable has range length 2, giving half-width 0.01517128, below $\Delta/2 = 0.015625$. Each secondary family uses $\alpha_{\text{each}} = 0.025$; E_{INFO} , B_{INFO} and B_{NONINFO} have range length 2 and half-width 0.01451463, and E_{NONINFO} has range length 1 and half-width 0.00725731. The three Study 002 families are controlled separately from one another and from the Study 001 families.

The frozen primary decision order was: (1) failure of any deterministic identity, coordinate, query-count, saved-output or audit check gives INVALID; (2) if either gate interval is not wholly inside $(-1/32, +1/32)$, the result is START-DEPENDENT and the primary unresolved; (3) a lower bound of Δ_P above 1/32 gives a meaningful positive directional-information effect relative to NONINFO; (4) an upper bound below -1/32 gives a meaningful adverse effect; (5) an upper bound at or below 1/32 gives BOUNDED BELOW THE POSITIVE ONE-BIT SCALE, with a separate statement of whether the interval lies wholly inside $(-1/32, +1/32)$; (6) otherwise UNRESOLVED. Each secondary estimand was classified against the same scale. Even a positive primary would have established a benefit relative to no-memory use only if the lower bound of B_{INFO} also exceeded 1/32. No sample extension, parameter change or follow-up cohort follows any branch.

3. Results

3.1 Study 001: execution, identity checks and record audit

The single production job (53530597) completed all 332,800 paths with exit code 0:0 and an elapsed time of 00:01:10. It made exactly 10,905,190,400 objective queries, 128 per update over 85,196,800 path-updates, with no N1, N2 or query-count failure. The frozen analyzer (job 53530688) and the separately implemented internal audit job (53530689) both completed with exit code 0:0. The accepted record lists 0 post-outcome changes.

Two audit tools, written separately from the producer and importing none of its code, checked the saved outputs. A C++ replay tool (MMEM-AUDIT-REPLAY-1.0.1) regenerated blocks 0-63 from block identifiers and keyed random streams and compared 64 blocks, 512 path records, 131,072 update records and 16,777,216 candidate-level audit rows against the saved outputs. After passing known-answer tests for SHA-256, Random123 Philox4x32-10 and multiply-high rejection, it reported PASS with zero mismatches. A Python verifier (MMEM-AUDIT-RECORDS-1.2.0) reconstructed all 22 saved estimates from saved path outputs, confirmed both SHAM identities in all 41,600 blocks, checked 332,800 path records and 85,196,800 update records, matched 97 manifest-listed files, recomputed the six frozen half-widths, and reported PASS with zero mismatches. These audits establish internal reproducibility of the recorded run. They were carried out within the project, partly with AI assistance; they are not external peer review and not an independent scientific replication.

3.2 Study 001: the start-state gate passed

Under HALF, the late-window ACTIVE frequency from ALL_M exceeded that from ALL_F by $D_{\text{HALF}} = 0.000723$, with familywise interval $[-0.014898, 0.016343]$. Under ZERO, $D_{\text{ZERO}} = -0.000096$, with interval $[-0.015717, 0.015524]$. Both intervals lie wholly inside $(-1/32, +1/32)$; in percentage points, the two start contrasts were $+0.072$ and -0.010 . Rule 2 therefore did not apply, and the enrichment rules were evaluated.

3.3 Study 001: recurrence strongly enriched the memory-use allele

The start-averaged late-window ACTIVE memory-use frequency was 0.810292 under HALF and 0.368657 under ZERO (Table 2, Figure 2). Relative to the exact neutral expectation, $C_{\text{abs}} = 0.310292$, with familywise interval $[0.302482, 0.318102]$: under recurrence the allele exceeded the neutral 1/2 baseline by 31.029 percentage points. Relative to the nonrecurrent law, $C_{\text{rec}} = 0.441635$, with interval $[0.426015, 0.457256]$: recurrence raised the late-window frequency by 44.164 percentage points. Both lower bounds exceed $\Delta = 0.03125$ by a wide margin. Decision rule 3 applied, and the frozen primary decision is **SUPPORTS RECURRENCE-ATTRIBUTABLE SELECTIVE ENRICHMENT**.

The individual cell means show the same pattern from both starts. Under HALF, ACTIVE frequencies were 0.809931 from ALL_F and 0.810654 from ALL_M. Under ZERO they were 0.368705 and 0.368609, below the neutral 1/2; this ZERO deficit is descriptive, because no inferential bound on $A_{\text{ZERO}} - 1/2$ was frozen. SHAM frequencies were 0.500083 and 0.499917 under ZERO and 0.499089 and 0.500911 under HALF; each pair sums to 1, as identity N2 requires.

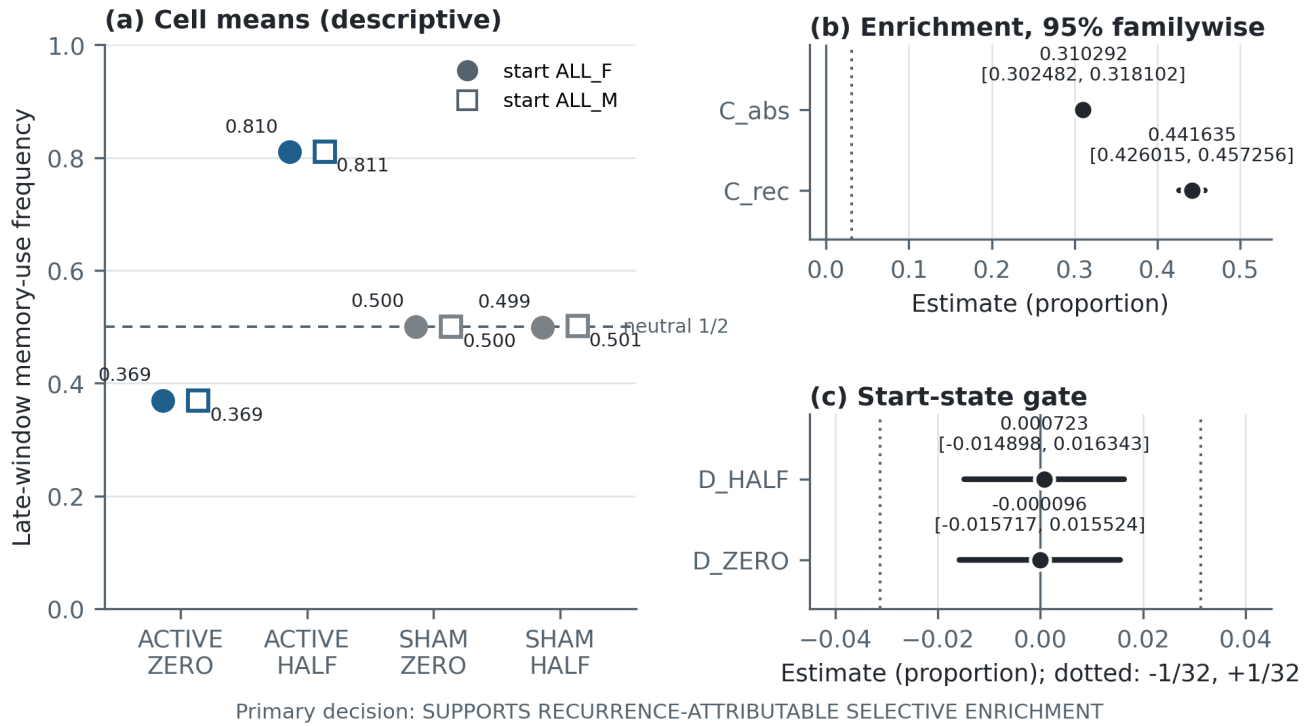


Figure 2. Study 001 allele-frequency outcomes. (a) Late-window memory-use frequency in all eight cells; markers distinguish the ALL_F and ALL_M starts, and the dashed line marks the exact neutral start-averaged expectation of 1/2. These cell means are descriptive. (b) The two enrichment estimands, C_abs and C_rec, and (c) the two start-state contrasts, D_HALF and D_ZERO, each with its frozen 95% familywise Hoeffding-Bonferroni interval; dotted lines mark the one-individual scale (+1/32 in b; -1/32 and +1/32 in c). Both enrichment estimands have lower bounds above +1/32, and both start-state contrasts lie inside the gate. All plotted values are read from the saved analyzer records at figure-generation time.

3.4 Study 001: population accuracy showed no resolved one-bit benefit

Under HALF, start-averaged ACTIVE accuracy was 0.628395, compared with 0.619018 for SHAM (Table 2, Figure 3). The recurrent-law accuracy contrast P_{abs} was 0.009377, with familywise interval [-0.005138, 0.023892]: an estimated contrast of +0.938 percentage points whose upper bound, 2.389 points, lies below the prespecified 3.125-point one-correct-bit threshold. The frozen classification is **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE**. Because this interval also includes zero, even the sign of the recurrent-law accuracy contrast is not resolved at the familywise level.

The recurrence interaction P_{rec} was 0.011360, with interval [-0.017669, 0.040390], or +1.136 percentage points with an interval from -1.767 to +4.039 points. The interval spans both zero and the one-bit scale, and the frozen classification is **UNRESOLVED**. The P_{rec} half-width, 0.02902925, is only slightly below Delta_P, so this design had limited capacity to resolve the interaction at the one-bit scale; the half-width was fixed in the frozen specification.

Under ZERO, start-averaged ACTIVE accuracy was 0.605327, compared with 0.607310 for SHAM: a small adverse descriptive contrast of -0.001983, or -0.198 percentage points. It carries no inferential bound and is retained as reported. SHAM accuracies were identical between starts (0.607310 under ZERO and 0.619018 under HALF), as identity N1 requires. The analyzer recorded the descriptive relation of P_{abs} to allele enrichment as DISAGREEMENT and that of P_{rec} as UNCERTAINTY.

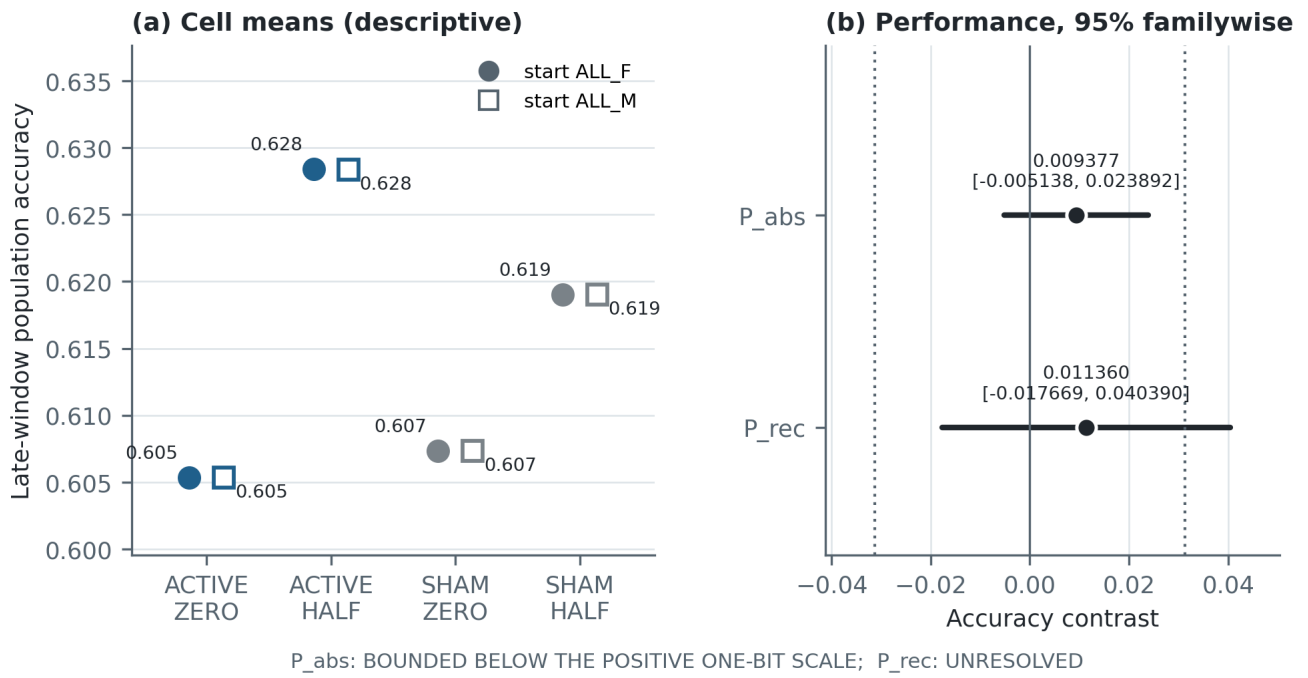


Figure 3. Study 001 population-accuracy outcomes. (a) Late-window mean population accuracy in all eight cells (descriptive); SHAM accuracies coincide between starts by identity N1. (b) The two secondary estimands with their frozen 95% familywise Hoeffding-Bonferroni intervals; dotted lines mark the one-correct-bit scale ($-1/32$ and $+1/32$). P_abs is bounded below the positive one-bit scale and P_rec is unresolved. All plotted values are read from the saved analyzer records at figure-generation time.

Table 1. Study 001 inferential estimands and frozen classifications. Intervals are two-sided 95% familywise Hoeffding-Bonferroni intervals over 41,600 blocks, computed separately for the primary and secondary families.

Estima nd	Estimate	95% familywise interval	Half-width	Frozen classification
C_abs	0.31029 2	[0.302482, 0.318102]	0.007810 23	Lower bound exceeds 1/32
C_rec	0.44163 5	[0.426015, 0.457256]	0.015620 46	Lower bound exceeds 1/32
D_HALF	0.00072 3	[-0.014898, 0.016343]	0.015620 46	Inside (-1/32, +1/32)
D_ZER O	-0.00009 6	[-0.015717, 0.015524]	0.015620 46	Inside (-1/32, +1/32)
P_abs	0.00937 7	[-0.005138, 0.023892]	0.014514 63	BOUNDED BELOW THE POSITIVE ONE-BIT SCALE
P_rec	0.01136 0	[-0.017669, 0.040390]	0.029029 25	UNRESOLVED

Table 2. Study 001 descriptive late-window cell means (no inferential bounds were frozen for these 16 records).

Cell	Memory-use frequency	Population accuracy
ACTIVE, ZERO, ALL_F	0.368705	0.605324

Cell	Memory-use frequency	Population accuracy
ACTIVE, ZERO, ALL_M	0.368609	0.605330
ACTIVE, HALF, ALL_F	0.809931	0.628405
ACTIVE, HALF, ALL_M	0.810654	0.628385
SHAM, ZERO, ALL_F	0.500083	0.607310
SHAM, ZERO, ALL_M	0.499917	0.607310
SHAM, HALF, ALL_F	0.499089	0.619018
SHAM, HALF, ALL_M	0.500911	0.619018

3.5 Study 001: the joint result

Taken together, the primary and secondary families give one result rather than two. Lag-two recurrence strongly enriched a mutation-generated memory-use allele despite its fixed cost of one fresh proposal, both late-window start contrasts passed the prespecified finite-horizon $\pm 1/32$ start-state gate, and the enrichment was attributable to recurrence at the one-individual scale. Over the same paths, the recurrent-law ACTIVE-SHAM accuracy contrast was bounded below one correct bit per survivor with sign unresolved, and whether recurrence changed the accuracy effect of the allele was unresolved. In this model, selection on the use of a supplied cache was strong, while the ACTIVE-SHAM accuracy contrast was small at most and was not resolved at the prespecified scale.

3.6 Study 002: execution and record audit

The Study 002 production run (job 53553084) completed all 249,600 paths and made exactly 8,178,892,800 objective queries, 128 per update over 63,897,600 path-updates. The frozen analyzer (job 53553256) saved the 19 prespecified estimate records and the decision record, and a separately implemented audit (job 53553294) checked them. A C++ replay tool (PCONV-AUDIT-REPLAY-1.0.0) regenerated blocks 0-63 and compared 384 path records, 98,304 update records, 12,582,912 candidate-level audit rows and 16,384 permutation records with the saved outputs. Its known-answer tests included the Fisher-Yates and rejection-sampling construction, distance matching of the decoy at every displacement weight from 0 to 32, the arm-specific probe rule and the HALF law; it reported PASS with zero mismatches. A Python verifier (PCONV-AUDIT-RECORDS-1.0.0), which generated no random draws, reconstructed all 19 saved estimates, confirmed both SHAM identities in all 41,600 blocks, checked 249,600 path records and 63,897,600 update records with 128 queries at every path-update, matched 98 manifest-listed files and recomputed the seven frozen half-widths; it reported PASS with zero mismatches. As for Study 001, these are internal reproducibility audits, not external peer review or independent scientific replication.

An earlier Study 002 production submission failed before any computation because a required command-line argument, --design-go, was missing. It exited with status 64 and produced no production output, random draw, model execution or outcome. A prospective recovery addendum was recorded before the corrected run. This infrastructure-only event is documented in Supplementary Section S24.

3.7 Study 002: the start-state gate passed

D_INFO = 0.000129, with familywise interval [-0.015042, 0.015301], and D_NONINFO = -0.001268, with interval [-0.016439, 0.013903]; in percentage points the two start contrasts were +0.013 and -0.127. Both intervals lie wholly inside $(-1/32, +1/32)$, so rule 2 did not apply.

3.8 Study 002: directional information had a bounded total effect on population accuracy

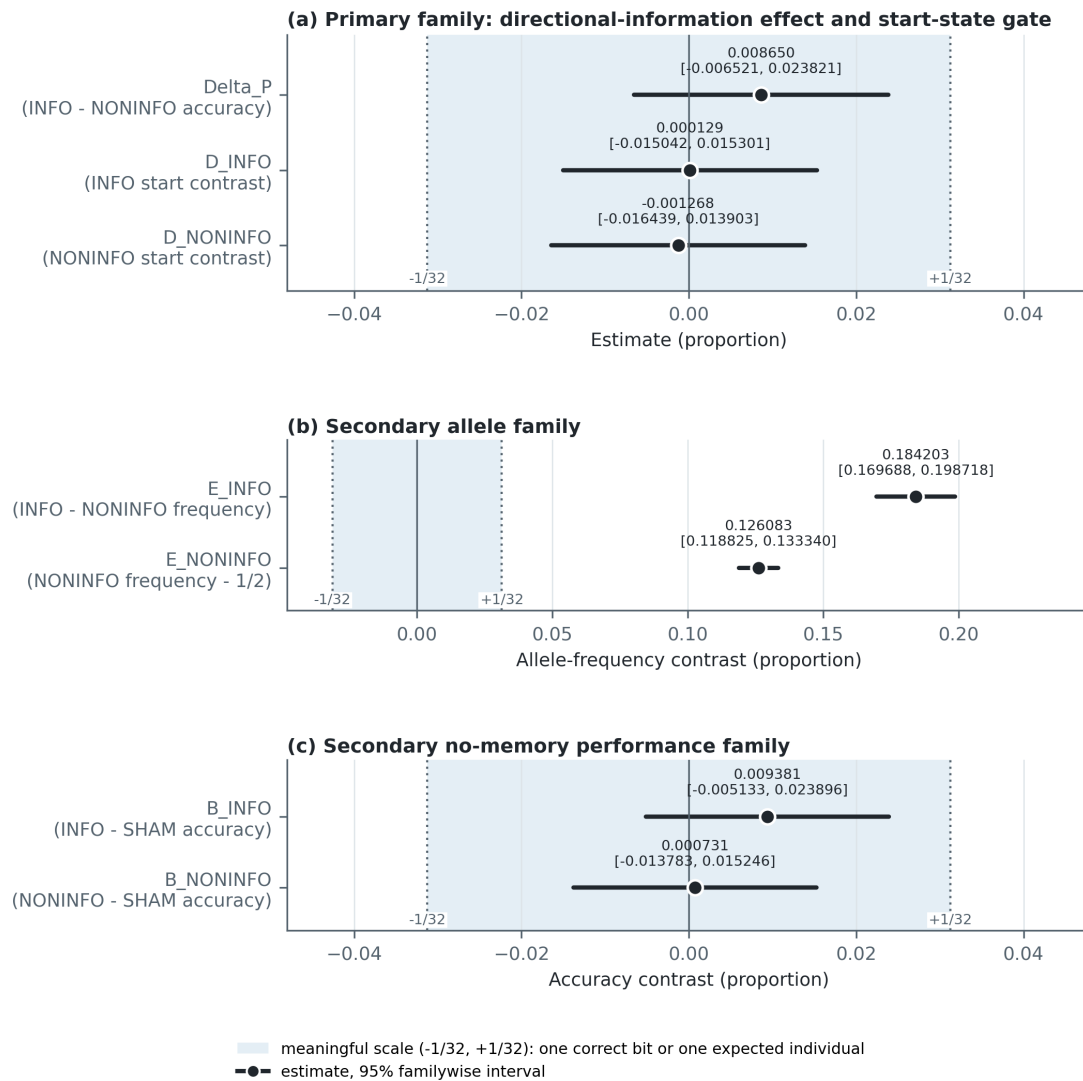
Start-averaged late-window accuracy was 0.628390 in INFO and 0.619740 in NONINFO (Table 4). The frozen primary estimate was $\Delta_P = 0.008650$ (0.00864996836735652 at saved precision), with familywise interval [-0.006521, 0.023821] ([-0.006521316095375593, 0.023821252830088634] at saved precision). Directional cache information therefore changed late-window population accuracy by an estimated +0.865 percentage points, with an interval from -0.652 to +2.382 points. The upper bound lies below the 3.125-point one-correct-bit scale, the whole interval lies inside $(-1/32, +1/32)$, and the interval contains zero. Decision rule 5 applied, and the frozen primary decision is **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE**; the sign of the effect is unresolved.

3.9 Study 002: directional information strongly amplified selection without a resolved absolute performance gain

Directional information strongly changed allele selection (Figure 4). Start-averaged late-window memory-use frequency was 0.810286 in INFO and 0.626083 in NONINFO, and exactly 1/2 in SHAM in every block. $E_{\text{INFO}} = 0.184203$, with familywise interval [0.169688, 0.198718]: directional retrieval left 18.420 percentage points (16.969 to 19.872) more memory-use carriers than the matched control. $E_{\text{NONINFO}} = 0.126083$, with interval [0.118825, 0.133340]: the matched control itself remained 12.608 points (11.883 to 13.334) above neutrality. Both are classified **MEANINGFUL POSITIVE AT THE ONE-BIT SCALE**.

Performance relative to the label-blind SHAM reference, whose start-averaged accuracy was 0.619009, did not follow. $B_{\text{INFO}} = 0.009381$, with interval [-0.005133, 0.023896], or +0.938 points (-0.513 to +2.390); $B_{\text{NONINFO}} = 0.000731$, with interval [-0.013783, 0.015246], or +0.073 points (-1.378 to +1.525). Both are classified **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE**, and neither sign is resolved. Because the lower bound of B_{INFO} does not exceed 1/32, Study 002 does not demonstrate a one-bit benefit of directional retrieval relative to no-memory use. The analyzer recorded the prespecified interpretive branch: "Directional information selects for policy use without converting into one-bit population performance, sharpening the selection-performance separation." These secondary classifications cannot alter the primary decision.

The cell means show the same pattern from both starts. INFO frequencies were 0.810221 from ALL_F and 0.810351 from ALL_M, and NONINFO frequencies 0.626717 and 0.625449. SHAM frequencies were 0.499678 and 0.500322, whose exact fractions sum to 1 as identity N2 requires, and SHAM accuracy was 0.619009 from both starts, as N1 requires.



Primary decision (rule 5): BOUNDED BELOW THE POSITIVE ONE-BIT SCALE; interval wholly inside (-1/32, +1/32): yes

Figure 4. Study 002 directional-information outcomes. (a) The primary family: the directional-information effect on population accuracy, Delta_P (INFO minus NONINFO), and the two start-state contrasts D_INFO and D_NONINFO. (b) The secondary allele family: E_INFO (INFO minus NONINFO memory-use frequency) and E_NONINFO (NONINFO minus the exact neutral 1/2). (c) The secondary no-memory performance family: B_INFO and B_NONINFO, accuracy relative to SHAM. Each estimate is shown with its frozen 95% familywise Hoeffding-Bonferroni interval; the shaded band and dotted lines mark the meaningful scale (-1/32, +1/32), one correct bit in accuracy or one expected individual in allele frequency. Delta_P lies wholly inside the band, both allele contrasts lie far above it, and both performance contrasts have upper bounds inside it. All plotted values are read from the saved Study 002 analyzer records at figure-generation time.

Table 3. Study 002 inferential estimands and frozen classifications. Intervals are two-sided 95% familywise Hoeffding-Bonferroni intervals over 41,600 blocks, computed separately for the primary, secondary allele and secondary performance families.

Estim and	Family	Estimate	95% familywise interval	Half-width	Frozen classification
Delta_P	Primary	0.008650	[-0.006521, 0.023821]	0.01517128	BOUNDED BELOW THE POSITIVE ONE-BIT SCALE (rule 5; wholly inside $\pm 1/32$)
D_INFO	Primary (gate)	0.000129	[-0.015042, 0.015301]	0.01517128	Inside (-1/32, +1/32)

Estim and	Family	Estimate	95% familywise interval	Half-width	Frozen classification
D_NO NINFO	Primary (gate)	-0.001268	[-0.016439, 0.013903]	0.01517128	Inside (-1/32, +1/32)
E_INF O	Secondary allele	0.184203	[0.169688, 0.198718]	0.01451463	MEANINGFUL POSITIVE AT THE ONE-BIT SCALE
E_NO NINFO	Secondary allele	0.126083	[0.118825, 0.133340]	0.00725731	MEANINGFUL POSITIVE AT THE ONE-BIT SCALE
B_INF O	Secondary performance	0.009381	[-0.005133, 0.023896]	0.01451463	BOUNDED BELOW THE POSITIVE ONE-BIT SCALE
B_NO NINFO	Secondary performance	0.000731	[-0.013783, 0.015246]	0.01451463	BOUNDED BELOW THE POSITIVE ONE-BIT SCALE

Table 4. Study 002 descriptive late-window cell means (no inferential bounds were frozen for these 12 records) and their start averages.

Cell	Memory-use frequency	Population accuracy
INFO, ALL_F	0.810221	0.628355
INFO, ALL_M	0.810351	0.628425
NONINFO, ALL_F	0.626717	0.619744
NONINFO, ALL_M	0.625449	0.619737
SHAM, ALL_F	0.499678	0.619009
SHAM, ALL_M	0.500322	0.619009
INFO, start average	0.810286	0.628390
NONINFO, start average	0.626083	0.619740
SHAM, start average	1/2 (exact in every block)	0.619009

3.10 The two-study result

The two prospective experiments give one paper-level result. In Study 001, lag-two recurrence strongly selected a costly memory-use policy while the recurrent-law accuracy contrast with SHAM was bounded below one correct bit with sign unresolved. In Study 002, under the same aligned law, directional cache information strongly amplified selection on the same policy: relative to a control that paid the same displacement cost and placed its candidate at the same distance from the parent, directional retrieval left 18.420 percentage points more memory-use carriers. Yet the total population-accuracy effect of that same information, mediated through evolved policy use, was bounded below the one-bit scale with sign unresolved, and neither INFO nor NONINFO showed a resolved accuracy gain over SHAM. Directional information therefore strongly amplified selection on the memory-use policy without a resolved conversion into population performance at the prespecified scale.

The 22 Study 001 records and the 19 Study 002 records are separate accepted estimate sets, 41 saved estimate records in total, from separate cohorts with separately controlled familywise families; they are not pooled. The Study 002 INFO arm applies the same retrieval rule as the Study 001 ACTIVE-HALF cells, and its descriptive start-averaged frequency and accuracy (0.810286 and 0.628390) are close to the Study 001 values (0.810292 and 0.628395). Study 002 extends the Study 001 producer and was neither designed nor analyzed as an independent replication, so this agreement is descriptive only.

4. Discussion

The central observation is a separation between what selection favored and what the population gained. In Study 001, under the aligned recurrent law, more than four-fifths of late-window survivors carried the memory-use allele in both start conditions over the 256-update horizon, whereas the ACTIVE-SHAM accuracy contrast, in which SHAM labels are inert, was bounded below one additional correct bit per survivor. Under the nonrecurrent law, where the cache carried no lag-two information, the allele fell descriptively below its neutral frequency, and ACTIVE accuracy was descriptively slightly below SHAM.

Study 001 was not designed to identify the mechanism of this separation, and the following explanations are hypotheses for future work rather than findings. Survival depends on the relative scores of candidates within an update. A carrier whose cache matches a recurrent target supplies a candidate that outscores most fresh proposals, so carriers can be strongly overrepresented among survivors even if the survivor set, which is dominated by the best candidates from all families, improves only modestly on average. At non-recurrent updates the same carriers have given up one fresh draw, and the local child of every parent already supplies a near-copy of the best current candidate, which may limit what a retrieved record adds to the population's mean fit. The ZERO accuracy contrast is in the direction expected from displaced exploration, but it is small and descriptive.

Study 002 sharpens this separation rather than reversing it. Its control removes the direction of the retrieved displacement at recurrent updates while keeping the same displacement cost and the same distance from a parent that has just survived selection. Relative to that control, directional information added a large and well-resolved selective advantage to the memory-use policy, but the total population-accuracy effect of the same information was bounded between -0.652 and +2.382 percentage points. Two further observations are descriptive and motivate hypotheses only. First, NONINFO was itself selected well above neutrality; one possibility is that a candidate placed at the cache's distance from a recently selected parent tends to score better than a uniform fresh proposal even without directional information, particularly because a large share of displacements had weight 0 and so reproduced the parent (Supplementary Section S23). Second, neither arm showed a resolved accuracy gain over SHAM, consistent with the Study 001 mechanisms suggested above, in which strong relative advantage among candidates can coexist with a small change in the survivors' mean fit. Study 002 was not designed to test these explanations.

These results are consistent in spirit with prior findings that the marginal benefit of learning or memory can be small even where memory is favored [4], and with models in which memory-related loci or anticipatory effects evolve only under particular environmental regimes [5, 7]. The present design adds an exact neutral benchmark for a mutable use allele, a cost expressed in the same currency as search itself, and prospectively separate scales for enrichment and performance. Its main methodological implication is that an allele-frequency response to environmental structure, however strong, should not be read as a population-performance

response without a separately stated performance estimand and scale. Study 002 extends that implication to the information content of the retrieved record itself: even an allele-frequency effect attributable to directional information, measured against a control matched on displacement cost and parent distance, need not convert into a commensurate population-performance effect.

The result is deliberately narrow. HALF matches the recurrence lag to the retrieval delay and so favors retrieval by construction; no other lag or copy probability was tested. The opportunity cost is specific to replacing one effectively uniform proposal in a four-candidate budget. The cache architecture was supplied rather than evolved. The Study 002 control matches parent distance and one common coordinate relabeling, not every relation among genotypes, and its primary estimand is a total effect through evolved policy use rather than a per-use value of information. Within those conditions, the allele inferences are strong and the performance inferences are bounded or unresolved; none should be generalized beyond them.

5. Limitations

- **One aligned recurrent law.** HALF was a prior, outcome-informed choice that matches the cache delay by construction, and Study 002 used HALF alone. No other lag, copy probability or target law was tested, and the result does not establish selection under arbitrary recurrence.
- **One fixed cost.** The cost is the replacement of one of two effectively uniform fresh proposals. It is a model-specific search-budget trade-off, not a calibrated energetic or biological cost and not a general cost scale. No other cost was tested.
- **One mutation rate, one horizon, one model class.** Both studies use mutation rate 1/32, 256 updates and one abstract model class; Study 002 extends the Study 001 producer and is not an independent model-class reproduction.
- **Supplied architecture.** The one-record private cache and its retrieval rule were supplied. Only the use policy mutated. Neither study addresses the spontaneous origin of memory architecture or capacity.
- **Retention and retrieval are not separated.** The allele jointly determines whether the one-step record is retained and whether it is retrieved; every individual has a cache slot, but only M carriers receive a valid record, so this design cannot separate selection on retention from selection on retrieval.
- **A total, not per-use, information effect.** Delta_P is the total population-accuracy effect of directional cache information beyond the parent and matched displacement length, mediated through evolved policy use. INFO and NONINFO diverge in policy frequency, cache use and survivors, so Delta_P is not a per-use value-of-information estimate and supports no per-use causal claim.
- **Control geometry.** The shared permutation preserves each decoy's distance to its parent and one common coordinate relabeling of displacement masks, but not every cross-genotype relation. Displacements of weight 0 or 32 cannot be scrambled and retain direction. In NONINFO, 42,175,734 of 100,158,252 decoy applications from ALL_F and 45,696,359 of 108,736,742 from ALL_M had weight 0 (42.1% and 42.0%); none had weight 32.
- **Finite horizon; start-state gate only.** In both studies, both late-window start contrasts were bounded within one expected individual (95% familywise) over 256 updates. This is not a demonstration of general start-independence, stationarity, equilibrium, evolutionary stability, rare-mutant invasion or fixation probability.
- **Performance not resolved at the one-bit scale.** P_abs excludes a gain of one correct bit but does not resolve its sign; P_rec is unresolved, and its half-width is close to the one-bit

scale. In Study 002, Delta_P, B_INFO and B_NONINFO exclude a positive one-bit effect but none resolves its sign. Population benefit was not demonstrated in either study. The small adverse ZERO accuracy contrast and the ZERO allele deficit are descriptive.

- **Separate families and separate estimate sets.** Each of the two Study 001 families and the three Study 002 families has 95% familywise coverage; coverage is not controlled jointly across families, across studies or across the 28 descriptive means. The 22 Study 001 and 19 Study 002 records are separate accepted estimate sets and are not pooled or presented as independent replications.
- **What SHAM does not control.** SHAM is an exact neutral benchmark for label dynamics, not a control for alternative caches, alternative costs or alternative retrieval rules. ACTIVE and SHAM genotypes and genealogies diverge at the first valid-M retrieval, so ACTIVE-SHAM contrasts are arm-level comparisons, not path-level counterfactuals; the same holds for INFO, NONINFO and SHAM in Study 002.
- **Abstract model.** The genotype, objective, selection rule and population are abstract. No biological, ecological, neural or engineering extrapolation and no claim of general population utility is supported, and the study does not replicate earlier internal private-memory studies.
- **Internal audit only.** Replay and record verification were performed by separately written tools within the project, partly with AI assistance. They are not external peer review or an independent scientific replication. Producer and audit tools were both written with AI assistance from the same specification; the audits detect implementation divergence, not a shared misreading of the specification.
- **No follow-up variants.** Neither study's specification permits a numerical extension, parameter grid, cost sweep, lag variant or new cohort in response to its outcome, and none is proposed here.
- **Bounded literature search.** The literature comparison rests on a structured search of open scholarly metadata with targeted citation chaining, not an exhaustive systematic review; it supports a bounded distinct-contribution claim only and cannot establish a unique nearest precedent. The search archive will be included in the public package (Section 7).

6. Methods summary

For Study 001, the model, cohorts and analysis follow the frozen specification PHASE2-MUTABLE-MEMORY-001 revision 1 exactly; Supplementary Sections S2-S10 give the full update order, target laws, allele mutation, cache inheritance and use, candidate budget, selection, estimands, interval construction and decision rules. Random values were drawn from canonical Random123 Philox4x32-10 streams whose keys are derived from SHA-256 of a purpose-specific namespace string and whose counters are the coordinates (block, update, entity, subindex); a coordinate-schema enumeration and collision audit preceded production. Ten deterministic fixtures, including hand traces, memory-better and memory-worse cases, duplicate and invalid-cache cases, full SHAM identity paths, target-law pairing, an exact weighted-selection fixture with a forced rejection, random-coordinate invariance and single-path regeneration, passed before production; none contains a sampled scientific outcome.

Per-update binary records were saved for every path, per-path late-window summaries and final-state hashes for every path, and full candidate-level rows for the fixed audit blocks 0-63. The frozen analyzer computed the 22 estimate records, their bounds and classifications from saved path outputs. Interval endpoints use exact rational estimates and high-precision half-widths. Descriptive diagnostics (mutation counts, cache-probe use and survival, and occupancy counters) were saved for interpretation only (Supplementary Section S12).

Study 002 follows the frozen specification PHASE2-PERFORMANCE-CONVERSION-002 revision 1 exactly; Supplementary Sections S16-S26 give the arm definitions, shared-permutation construction, event order, estimands, interval families, frozen decision order, all 19 records, diagnostics and audit. Random values come from the same Philox construction under the new namespace PHASE2-PERFORMANCE-CONVERSION-002|production-r1|<purpose>, with one added purpose, DECOY_PERMUTATION, generated for every block and update regardless of branch or arm. The specification required ten deterministic fixtures before production, covering bit-identity of the unchanged operators, Fisher-Yates and rejection fixtures with forced retries, INFO-NONINFO identity through innovation updates, distance matching at all 33 displacement weights, preservation of weights and overlaps under the shared permutation, the uniform subset law of the decoy, full SHAM identity paths, random-coordinate invariance and single-path regeneration; none contains a sampled production outcome. Per-update records, per-path summaries and final hashes, the 19 estimate records and full candidate rows and permutations for audit blocks 0-63 were saved. The frozen analyzer computed all estimates from saved path outputs in exact rational arithmetic with high-precision half-widths.

This is an abstract computational model. It involves no human participants, no animals and no personal data.

7. Data and code availability

A public reproducibility package is being prepared. It will contain the frozen specification, source code, configuration, build environment and SHA-256 manifests; the saved per-update, per-path and per-block outputs; candidate-level audit rows; analyzer records; internal audit receipts; the prospective design-review, production-authorization, production-submission and acceptance records; the literature-search archive (search script, API responses, deduplicated records and title screen); and this manuscript source with its figure and statistics-checking scripts. Manuscript text, figures and data will be released under the Creative Commons Attribution 4.0 International license (CC BY 4.0), and original code under the MIT License. No DOI, repository address or release date has been assigned yet. The authenticated outputs used here have the following SHA-256 values: decision record

5b3aea3c18c4530b2be9f29d5c64fdd8173b6b323823bf9cc3361bd4e7010fc0; 22 estimate records 83828a49936bb8364813e2634f11cb28c4de2212b4b162d96e616ddbc7b4f8c0;

descriptive diagnostics

51554620a5e302416edf8de8a073c6962abbc2c8967da80b91b83e4c3d5b685c; replay receipt 9b40322d0d09b90f0a6862533c1efdc7aff21bd699a882b9b9555c5f7c4b4857; record-verification receipt afd96dc0a5b190401c035fc66c8af8fd4d4603e7185d05bff73fb05e698cff5a; production inventory audit 622fa9e01a40c44ba545926e54b06e102d20b7fcf9e9c8e645ddaf9f7212e09c.

Prospective timing is recorded as follows (all times UTC, 2026-10-01). The frozen specification, design/PHASE2-MUTABLE-MEMORY-001-REV1.md (revision 1), has SHA-256 c17a3a1e9ac9cf2260f6743eaf08e4c2808080a767af16d2f9e9d2b9ad3e821d. Its terminal design review by a non-implementing AI reviewer (record CLAUDE-003, verdict GO) was completed at 01:55:25; the review has SHA-256

787cf32a20312f2682c761243f3dec04c5eb8036d14d6efc444e9830db37960a and its receipt 6277757f9ab3dfa611515c161776c17b561738eba6fa478ae0f8634e13c03176. The production execution authority (SHA-256

ee3cb68bed5d212e6ea10ece542e2e90c32cb9cea07d85fceda10fc56b375b1d) was issued at 12:53:04, and the sole production job (53530597) was submitted at 12:55:19 with no automatic retry and no parameter changes. The acceptance record has SHA-256

2eb3aaf5d754f4570a75522aa9e1f38358715c06c674d5a267732a1357926e6f. The frozen design

and its GO review therefore precede the only production authorization and submission.

Study 002 is a post-v1.0 prospective extension added in version 1.1.0 of this manuscript; the Study 001 values, records and provenance above are unchanged. The planned public package will also contain the Study 002 frozen specification, producer extension, configuration and manifests, saved per-update, per-path and per-block outputs, candidate-level audit rows and permutations for blocks 0-63, analyzer records, internal audit receipts, the record of the failed original production submission, the prospective recovery addendum and the acceptance record, under the same licenses. The authenticated Study 002 outputs have the following SHA-256 values: decision record 3f28b82d0cb40357a1068c464d1c5158c36ba8fbfe775026ce22c70113b3b84a; 19 estimate records 5efa213cdfeb985a20fd7bbc65fb36dd505c568670ad350b3223c2139eb0b68d; descriptive diagnostics 304843eb5d4a38761a6ce371c7cc125260041dfd910b29d5ba3cc377a6ac3553; replay receipt cbee79527876745e8918576f73a5d302377e4463d959318ca2f5feffa66416d4; record-verification receipt 09f5dccf4921c8d842ef60fb31aca712bbb582e87a2011c76c02efb658539197; production run manifest b7aceaf6638e8f06246c8fb9b182c280a8ed98c4444d93a4a60decdbafb80136; acceptance record fc7777239cd4442c35c01d6261fd71e6fbf7e7f6fd6e11ddcc5d24876ce2f4a2 (accepted at 07:10:51 UTC on 2026-10-02). The frozen Study 002 specification, design/PHASE2-PERFORMANCE-CONVERSION-002-REV1.md (revision 1), has SHA-256 58ee414d753fab042e6ccce1a4e6b95dd7f167633b4f33c1491687d9afcb895. Its non-implementing terminal review returned DESIGN_GO; the review has SHA-256 76b0c6172e61f60befecf183e83d9e43ea50c1b6bc1e93299668b76f55be60f, and DESIGN_002_GO.json (SHA-256 f4a6180109e0d212cbe784a7722a64576e5e7cb0fbfd4b23c757bb5e7ab84d6a) was frozen at 18:59:00 UTC on 2026-10-01. The original production authority (SHA-256 31c2100dd73de0090ded3059525aacd139565917086d2c19880b01e319c98992) was issued at 06:02:41 UTC on 2026-10-02. After the infrastructure-only failure, the failure record (SHA-256 289bf3188286476698136d499c12eb3a8c9e4e6fc5cde18fcb15718395f4afaa) was accepted at 06:28:45, the prospective recovery addendum (SHA-256 db12c0143b6ba2c0b8cd3771ad18b9afe12feb069fe2a97d15227f4fad836e83) was issued at 06:29:59, and the one-recovery execution authority (SHA-256 c4cb2e4c162df92a761b2fff391b6e81e000055db88d7cbaeb954b0750b4e085) was issued at 06:32:55, all UTC on 2026-10-02.

8. AI assistance

AI systems assisted with design criticism, implementation, code review, analysis checking, literature organization and drafting, under human direction. The author directed the work, made the scientific decisions and is responsible for the content. Internal reviews performed by AI models, including the non-implementing AI review of the frozen specification, are not external peer review. The integration of Study 002 into version 1.1.0 was also drafted with AI assistance.

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